Module 7: Assignment

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Questions

1. **(a) What are the maximum speeds that can be achieved by Bluetooth? (b) On what do they depend?**

Bluetooth can reach speeds of about 723 kbps in a perfect scenario. Real-life speeds may not reach this number due to environmental and situational factors. Things such as the number of users transmitting via Bluetooth and how much they are transmitting affect the speed.

2. **Describe the difference between the 'classic' Bluetooth and the Bluetooth low energy air interface?**

There are a few differences between classic Bluetooth and BLE regarding the air interface. One is that Bluetooth uses a faster frequency-hopping mechanism and 1 MHz channels. BLE uses slower frequency hopping and larger channels of 2 MHz. This and a couple of other differences allow BLE to have lower power consumption.

3. **How are data transferred over the BLE ATT protocol?**

Data that is transferred over the BLE ATT protocol follows the client-server model ("Att & Gatt: Data Representation and Exchange", 2024). The client makes requests to the server to get or set certain values on the server. The server then sends a response to the client to return the requested data or an acknowledgment message. Transferring data in this manner conserves power.

4. **How can several data streams for different applications be transferred simultaneously by the L2CAP protocol?**

Packets of different applications are given unique connection identifiers so that they can be told apart from each other and allow multiple connections at the same time. These IDs are given to connections during the establishment process.

5. How can several services use RFCOMM layer simultaneously?

BT Profiles must be registered in the Service discovery database in order to use the RFCOMM layer. Devices wanting to use registered profile services must query the database for the RFCOMM channel number.

6. What are the differences between the hands-free profile and the SIM-access-profile?

The hands-free profile sees the connected Bluetooth device as an extension of the phone, the connection is established with the phone, and the audio data is relayed to the Bluetooth device. The SIM-access profile reads the SIM card remotely as if inserted in the hands-free device and handles the connection itself. The transceiver on the phone is disabled for the duration of this connection. For this profile to work, it must have more complex hardware including a GSM/UMTS module.

7. What are the tasks of the link manager?

The link manager has various tasks. It establishes an SCO, eSCO, or ACL connection, configures the connection, carries out master-slave role change procedure, authentication, frequency hopping management, and switching to power saving mode.

8. **What are the tasks of the service discovery database?**

The service discovery database has information about exposed services hosted on a device devices can ask the database about available services and how to access them. This query happens during connection establishment.

9. **What is FHSS and which enhanced functionalities are available with Bluetooth 1.2 in this regard?**

Frequency Hopping Spread Spectrum is the mechanism that sends individual packets on different frequencies. This helps avoid staying on congested channels for a long period of time. Adaptive Frequency Hopping Spread Spectrum is a new feature in Bluetooth 1.2 that avoids channels that are known to be noisy.

10. **What is the difference between authentication and authorization?**

Authentication is the process of making sure that the device on the receiving end is who they say they are supposed to be. It verifies identity. Devices authenticate each other if they have been paired in the past. Authorization is making sure that devices cannot access services they do not have permission to access.

11. **What is the difference between inquiry and paging?**

Inquiry is when a device sends out an inquiry request and unknown devices will respond with its device ID. Paging is used to connect two devices that have already been paired in the past.

12. What kinds of power-saving mechanisms exist for Bluetooth devices?

Various mechanisms exist in Bluetooth to save power. Connection hold is when the device disables its transceiver for a certain period. Connection sniff is when a device disables its transceiver for a period but checks once in a while to see if there is data queued to be received. Connection park is when a device gives up its MAC address and disables its transceiver for a long period of time until checking if a connection needs to be re-established.

13. Which profiles can be used to quickly transfer files and objects between two Bluetooth devices?

The Object Exchange profile offers quick transfer of files and objects between two devices. This profile makes up other profiles including the Object Push Profile and the File Transfer Profile.

14. Why are such a high number of different Bluetooth profiles required?

Bluetooth is very universal, it can be used for many different applications and data transfer modes. Profiles help ensure that transmission in each of these scenarios works properly and does what it is meant to do. Profiles define communication behavior and mechanisms that devices must adhere to when exchanging data for a specific task.

References

Att & Gatt: Data Representation and Exchange. Nordic Developer Academy. (2024, May 22). https://academy.nordicsemi.com/courses/bluetooth-low-energy-fundamentals/lessons/lesson-1-bluetooth-low-energy-introduction/topic/att-gatt-data-representation-and-exchange/#:~:text=The%20ATT%20layer%20is%20the,the%20data%20from%20the%20server.